A Project Report on

GNN-BASED DYNAMIC USER PAIRING AND SLOT ASSIGNMENT FOR HYBRID NOMA NETWORKS

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Abstract

In modern wireless communication systems, the increasing number of users accessing the network simultaneously creates challenges in efficiently managing limited communication resources. Non-Orthogonal Multiple Access (NOMA) enables multiple users to share the same time and frequency resources, improving resource utilization and supporting higher network capacity. However, efficient user pairing and time-slot assignment remain challenging because of variations in channel conditions, interference, and user requirements. Improper pairing and resource allocation can reduce the achievable data rate, affect fairness, and increase the probability of communication failure. To address these challenges, this project presents a Hybrid Graph Neural Network (GNN)-Based Approach for User Assignment and Slot Pairing in NOMA Networks, combining graph-based learning, classical machine learning techniques, and wireless communication performance evaluation.

The proposed system is implemented in Python and uses simulated wireless network scenarios to evaluate user relationships and resource allocation strategies. Each user is represented as a node in a graph, with features such as user position, distance, channel gain, channel gain in decibels, signal-to-noise ratio (SNR), and minimum data-rate requirements. The GNN-based approach processes user relationships to support user pairing and slot assignment under the simulated network conditions. The system also incorporates Power-Domain NOMA (PD-NOMA) and Successive Interference Cancellation (SIC) for evaluating communication between paired users, along with a Sparse Code Multiple Access (SCMA) experiment to investigate its performance under different signal conditions. The hybrid scheduling approach is examined in relation to the implemented NOMA and SCMA components.

In addition to graph-based resource allocation, classical machine learning models, including Linear Regression, Ridge Regression, Lasso Regression, Decision Tree Regression, and Random Forest Regression, are evaluated for data-driven rate prediction. Their performance is compared using cross-validation and held-out test data, with metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). A paired bootstrap analysis is also used to estimate the uncertainty in the difference between prediction errors. These predictive modelling results are analyzed separately from the GNN's wireless communication performance to distinguish rate prediction accuracy from resource allocation effectiveness.

The wireless performance evaluation compares the GNN-based approach with conventional and reference methods, including Orthogonal Multiple Access (OMA), random pairing, near-far pairing, greedy allocation, and an optimization-based reference. Performance is assessed using metrics such as sum rate, average user rate, fairness, and

outage probability. The experiments use simulated data with separate training, validation, and test scenarios for predictive modelling, allowing the implemented methods to be evaluated systematically. The results demonstrate the trade-offs between the compared approaches and help identify the strengths and limitations of the proposed system under the tested conditions. Overall, this project demonstrates the application of Graph Neural Networks and classical machine learning to user pairing, slot assignment, and wireless resource allocation in simulated hybrid NOMA networks, while providing a comparative evaluation of prediction accuracy and communication performance.

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Chapter 1

Introduction

1.1 Background

The rapid growth of wireless communication applications has increased the demand for high data rates, efficient spectrum utilization, and reliable connectivity. Conventional wireless communication systems often allocate separate orthogonal resources to different users. Although this approach reduces interference, it can limit the number of users served within a given bandwidth and time interval. Non-Orthogonal Multiple Access (NOMA) is a promising technique that allows multiple users to share the same time-frequency resources. By enabling simultaneous transmission and employing suitable detection techniques, NOMA can improve resource utilization. However, its performance depends on several factors, including user channel conditions, power allocation, interference management, and the selection of users sharing resources. In Power-Domain NOMA (PD-NOMA), users are multiplexed using different power levels, and Successive Interference Cancellation (SIC) is employed at the receiver to separate their signals. Sparse Code Multiple Access (SCMA) uses sparse codebooks to support multiple users over shared resources, with Message Passing Algorithm (MPA)-based detection commonly used to recover transmitted information. Hybrid NOMA networks can combine these approaches to provide flexible resource allocation. However, deciding which users should share a resource and which time slot should be assigned to each group is a complex optimization problem. The number of possible allocations increases as the number of users and available slots grows. Graph Neural Networks (GNNs) provide a way to process relational information between users. By representing users as nodes and their relationships as edges, a GNN can learn graph-based representations that capture relevant interactions. This project investigates the use of an edge-aware GNN for user pairing and slot assignment in hybrid NOMA networks.

1.2 Problem Statement

Efficient user pairing and slot assignment are essential for improving the performance of hybrid NOMA networks. Users may have different channel gains, distances, signal-to-noise ratios, and minimum data-rate requirements. Selecting unsuitable user pairs can increase interference, reduce achievable throughput, or negatively affect service reliability.

Traditional allocation strategies, such as random pairing and fixed heuristics, may not adequately adapt to varying network conditions. Furthermore, assigning multiple groups to the same slot can violate resource constraints and create scheduling conflicts.

Therefore, the problem addressed in this project is to develop a graph-based learning approach that uses user and channel information to support dynamic user pairing and slot assignment, subject to resource constraints, and evaluate its performance against conventional allocation methods.

1.3 Objectives of the Project

The main objectives are:

To develop a graph representation of users and their channel relationships in a hybrid NOMA network.

To implement an edge-aware Graph Neural Network that generates user-pairing affinities and slot-preference scores.

To perform user pairing and capacity-aware slot assignment using the learned outputs.

To support hybrid scheduling involving PD-NOMA, SCMA, and OMA for unpaired users where applicable.

To compare the proposed approach with OMA, random pairing, Near-Far pairing, greedy pairing, and an optimization-based reference.

To evaluate the system using sum rate, fairness, outage, and relevant SCMA communication metrics.

1.4 Scope of the Project

The project focuses on simulation-based resource allocation in hybrid NOMA networks. It includes synthetic wireless scenario generation, graph construction, GNN training, user pairing, slot assignment, hybrid scheduling, and performance evaluation.

The system is implemented using Python and PyTorch. It also includes a separate SCMA simulation for studying BER, SER, and data-rate behaviour across different SNR values.

The present evaluation is based on simulated scenarios. Performance in real-world wireless deployments, very large networks, and changing physical environments requires additional investigation.

1.5 Organization of the Report

This report is organized into six chapters. Chapter 1 introduces the project, its problem statement, objectives, and scope. Chapter 2 describes the hybrid NOMA system and graph-based representation. Chapter 3 explains the proposed GNN methodology and allocation workflow. Chapter 4 presents the implementation and simulation setup. Chapter 5 discusses the experimental results and comparisons. Chapter 6 summarizes the conclusions and identifies possible future extensions.

Chapter 2

System Model

2.1 Overview of Hybrid NOMA Networks

The considered system contains a set of wireless users served using a limited number of communication slots. Each user has associated characteristics, such as position, distance from the transmitter, channel gain, SNR, and minimum-rate requirement.

The allocation system determines suitable user pairs and assigns them to available slots. Depending on the scheduling decision, a slot can support PD-NOMA or SCMA communication. Users who remain unpaired can be served using an orthogonal allocation, subject to the implemented scheduling rules.

The overall objective is to evaluate whether graph-based learning can support effective resource allocation under the constraints of the simulated network.

2.2 Power-Domain NOMA

Power-Domain NOMA allows multiple users to share the same time-frequency resource by transmitting their signals at different power levels. The received signal can be represented as

where (y) is the received signal, (h_k) is the channel coefficient of user (k), (P_k) is its transmission power, (x_k) is its transmitted symbol, and (n) represents noise.

At the receiver, Successive Interference Cancellation can be used to decode signals successively and remove detected interference. The achievable performance depends on channel conditions, power allocation, and decoding order.

In this project, PD-NOMA is one of the communication modes considered by the hybrid scheduling framework. The precise achievable-rate calculation follows the implemented channel model and evaluation code.

2.3 Sparse Code Multiple Access

SCMA is a non-orthogonal multiple-access technique in which user data symbols are mapped to multidimensional sparse codewords. Multiple users can share resources through code-domain multiplexing.

The receiver can use a Message Passing Algorithm to estimate transmitted symbols by exploiting the sparse factor-graph structure. SCMA performance is evaluated through BER, SER, and data rate under different SNR conditions.

The project includes a separate SCMA simulation that studies communication performance as the signal-to-noise ratio changes.

2.4 Graph Representation of Users

A graph is represented as

$$\mathcal{G} = (V, E) \tag{2.1}$$

where (V) is the set of user nodes and (E) is the set of edges describing relationships between users.

Each node represents a wireless user. Node features include the user coordinates, distance, channel gain, SNR, and minimum-rate requirement, as supported by the implementation. Edge features represent relationships between users that can help the model estimate pairing suitability.

The graph representation allows the GNN to process both individual user characteristics and relationships between potential communication partners.

2.5 Problem Formulation

Let (U) denote the set of users and (S) denote the set of available slots. The allocation process selects user pairs and assigns them to slots while respecting the implemented resource constraints.

A general sum-rate objective can be expressed as

$$R_{\text{sum}} = \sum_{k \in \mathcal{U}} R_k(\mathcal{A}) \tag{2.2}$$

Let $\mathcal{A}$ denote the allocation matrix (or feasible allocation space), and let $R_k(\mathcal{A})$represent the achievable rate of user k as a function of the allocation decision $\mathcal{A}$. The objective is to find the optimal allocation policy that maximizes the sum-rate across the entire set of users $\mathcal{U}$. The allocation must satisfy applicable constraints, including slot capacity and feasible user assignments. Fairness and outage are evaluated as additional performance metrics.

The implemented system uses GNN-generated pairing affinities and slot preferences, followed by allocation and conflict-resolution logic. This learning-based approach is

compared with alternative allocation strategies.

Chapter 3

Proposed Methodology

3.1 System Architecture

The proposed system consists of the following stages:

Network scenario generation.

User feature extraction.

Graph construction.

GNN training and inference.

User-pair selection.

Slot assignment and conflict resolution.

Hybrid communication scheduling.

Performance evaluation.

The input stage provides information about users and their channel conditions. The graph builder converts the information into a graph representation. The GNN processes the graph and generates scores used by the allocation stage. Finally, the selected allocation is evaluated using the simulated communication model.

3.2 Graph Construction and Feature Extraction

The graph builder converts network scenarios into graph-based model inputs. Each user is represented by a node whose features describe its position and communication characteristics.

The feature set includes:

User coordinates.

Distance-related information.

Channel gain in linear and decibel representations.

Signal-to-noise ratio.

Minimum-rate requirement.

Edges represent relationships between users and carry relevant edge features. This representation allows the model to learn which user combinations may be suitable for sharing

a communication resource.

The constructed graph is provided to the GNN for training or inference.

3.3 Graph Neural Network Model

The project uses an edge-aware GNN identified in the implementation as HybridNOMAGNN. The model processes node features and edge information to learn representations of the network.

These representations are used to produce pairing-affinity scores and slot-preference logits. Pairing affinities help identify candidate user pairs, while slot-preference scores guide the assignment of selected pairs to available slots.

The model is trained using generated scenarios and corresponding oracle-derived labels. Training and validation data are used to learn model parameters and identify the best validation checkpoint.

The saved checkpoint is named best_gnn_model.pt.

The GNN is the learning component of the allocation pipeline; the final schedule also depends on the pairing, slot-assignment, and conflict-resolution logic.

3.4 User Pairing and Slot Assignment

The allocation procedure uses the learned outputs to select user pairs and assign them to available slots.

The main steps are:

Receive the graph representation of the current network.

Run the trained GNN to obtain pairing affinities and slot-preference scores.

Select candidate user pairs using the implemented pairing procedure.

Determine suitable slot assignments from the slot-preference scores.

Resolve allocation conflicts according to the implemented capacity rules.

Schedule the selected communication mode.

Evaluate the resulting allocation.

The scheduling procedure limits each slot to one allocation in the recorded configuration. Unused slots can remain idle, while unpaired users may receive OMA single-user allocations under the implemented rules.

3.5 Hybrid Scheduling and Baseline Methods

The framework supports hybrid scheduling involving PD-NOMA, SCMA, and OMA for single-user allocations. The selected communication mode depends on the implemented scheduling logic.

The proposed method is compared against the following baselines:

OMA: Orthogonal resource allocation.

Random pairing: Pairing selected randomly.

Near-Far pairing: Pairing based on differences in user channel conditions.

Greedy pairing: Pairing using a heuristic selection procedure.

Optimization reference: A reference allocation obtained using the project's optimization procedure.

GNN pairing: Allocation based on GNN-generated pairing and slot-preference outputs.

These methods provide a basis for comparing throughput, fairness, and outage under the simulated conditions.

3.6 Performance Metrics

Sum rate

Sum rate represents the aggregate achievable communication rate across the scheduled users.

It is used to evaluate the overall throughput of an allocation.

Jain's fairness index

Jain's fairness index measures the distribution of rates among users:

where:

$J \in [1/K, 1]$ represents the fairness metric value. K is the total number of active users sharing the system resources. R_k is the achievable rate (or throughput) allocated to user k.

A value closer to 1 indicates a more balanced distribution of user rates.

Outage percentage

Outage percentage represents the proportion of evaluated users that fail to satisfy the applicable minimum-rate criterion, according to the implemented evaluation procedure.

Bit error rate and symbol error rate

BER measures the fraction of incorrectly received bits, while SER measures the fraction of incorrectly detected symbols. Both are used to study the reliability of SCMA

communication under different SNR conditions.

Chapter 4

Implementation and Simulation Setup

4.1 Software and Tools

The project uses the following technologies:

- **Python:** Main programming language.
- **PyTorch:** GNN implementation and training.
- **NumPy:** Numerical computations.
- **pandas:** Data handling and metric tables.
- **scikit-learn:** Supporting regression, preprocessing, and clustering experiments.
- **Matplotlib:** Generation of plots and result visualizations.
- **YAML and argparse:** Configuration and command-line execution.

4.2 Project Modules

The project is organized into multiple modules to separate network simulation, graph learning, allocation, and evaluation.

Module	Purpose
main.py	Main entry point and command-line execution
channel_model	Wireless channel and network modelling
data_pipeline	Scenario preparation and graph construction
gnn_model	Edge-aware GNN architecture
training	Dataset generation and GNN training
pairing	User-pair selection and slot-allocation logic

Module	Purpose
evaluation	Performance evaluation and benchmark comparison
classical_models	Separate rate-prediction and clustering experiments

The project supports execution modes for training, evaluation, inference, comparison, clustering, and retraining, as available in the implementation.

4.3 Dataset Generation and Training

The simulation framework supports synthetic network scenarios and CSV-based inputs. Synthetic scenarios are used to generate user characteristics and graph-based training examples.

The recorded experiment uses the following dataset split:

Dataset	Number of scenarios
Training	200
Validation	40
Testing	50
Total	290

The GNN is trained using the training scenarios. Validation data are used to track model performance and identify the best checkpoint. Test scenarios are used for final evaluation and comparison with baseline methods.

4.4 Simulation Parameters

The reported experiment uses the following settings:

Parameter	Value
Number of users	10
Number of slots	5
Random seed	42
Training epochs completed	25
Execution device	CPU
Data source	Synthetic
Total bandwidth	10 MHz
Carrier frequency	2.4 GHz
Channel fading model	Rayleigh
Path-loss exponent	3.5
Minimum rate threshold	100 kbps

These values describe the previously reported experiment. They should be cross-checked against the saved configuration and experiment logs before final submission.

4.5 Execution Workflow

The implementation follows this sequence:

Configure the network and simulation parameters.

Generate or load network scenarios.

Extract user and channel features.

Construct graph-based inputs.

Generate training labels using the implemented oracle procedure.

Train the GNN and select the best validation checkpoint.

Generate pairing and slot preferences during inference.

Apply allocation and scheduling constraints.

Evaluate the proposed method and the baselines.

Save performance metrics and figures.

The resulting output files are stored in the project's results directories, including metric tables and figures used for analysis.

Chapter 5

Results and Discussion

5.1 GNN Training Results

The recorded training summary reports that the GNN completed 25 epochs. The best validation loss was obtained at epoch 10.

Metric	Recorded value
Epochs completed	25
Best validation loss	1.5596
Best validation epoch	10
Checkpoint	best_gnn_model.pt

The best validation checkpoint is selected based on validation performance. The recorded run continued to its epoch limit without early stopping.

Insert the training and validation loss plot generated by the final experiment.

Figure 5.1: GNN training and validation loss curves.

5.2 User Pairing and Slot Assignment

The recorded example allocation contains five pairs for ten users across five available slots.

Pair	Users	Slot	Method
1	U3, U6	S5	SCMA

Pair	Users	Slot	Method
2	U1, U2	S4	PD-NOMA
3	U0, U8	S3	PD-NOMA
4	U5, U7	S1	PD-NOMA
5	U4, U9	S2	PD-NOMA

Each slot contains one pair in this example topology. The result illustrates how the pairing and slot-assignment stages produce a feasible allocation for the recorded scenario. This is one example allocation and should not be interpreted as the same pairing being selected for every test scenario.

5.3 Comparison with Baseline Methods

The recorded benchmark results provide a comparison of the proposed GNN-based allocation method with five alternatives.

Method	Sum rate (Mbps)	Jain's fairness	Outage (%)
GNN pairing	17.69	0.529	10.00
OMA	19.49	0.335	50.40
Random pairing	16.04	0.526	1.60
Near-Far pairing	16.29	0.508	2.20
Greedy pairing	16.75	0.505	11.00
Optimization reference	17.59	0.523	6.40

5.3.1 Discussion

The recorded GNN result has a mean sum rate of 17.69 Mbps, compared with 17.59 Mbps for the optimization reference. The difference is approximately 0.10 Mbps when

rounded to two decimal places.

OMA has the highest reported mean sum rate, at 19.49 Mbps. However, it also has a substantially higher reported outage percentage of 50.40% and a lower fairness index of 0.335.

The GNN achieves the highest Jain's fairness index among the listed methods, at 0.529. Random pairing has the lowest reported outage percentage, at 1.60%. Therefore, the results do not show that one method is best for every metric.

The comparison highlights the trade-off between aggregate throughput, fairness, and outage. The GNN offers competitive sum-rate performance and the highest measured fairness among the evaluated methods, while further investigation is required to improve reliability.

5.4 SCMA Performance under Different SNR Conditions

The standalone SCMA simulation evaluates BER, SER, and data rate across different SNR values.

SNR (dB)	BER	SER	Data rate (Mbps)
-5	0.4342	0.6400	3.95
0	0.3717	0.5433	9.56
5	0.2942	0.4200	18.06
10	0.1625	0.2450	26.24
15	0.0417	0.0600	31.39
20	0.0033	0.0050	33.75

5.4.1 Discussion

The results show that BER and SER decrease as SNR increases. At low SNR, noise has a greater effect on symbol detection, leading to higher error rates. At higher SNR, the received signal quality improves, reducing detection errors.

The reported data rate increases from 3.95 Mbps at -5 dB to 33.75 Mbps at 20 dB. These results characterize the separate SCMA simulation and should not be interpreted as direct measurements of the GNN pairing algorithm.

5.5 Discussion and Limitations

The experiments demonstrate that graph-based learning can be incorporated into the user-pairing and slot-assignment workflow. The recorded results indicate competitive sum-rate performance and favourable fairness compared with the selected allocation baselines.

However, the results are based on simulated network scenarios. Performance may differ under real channel conditions, larger user populations, different traffic requirements, and changing network topologies.

The reported outage percentage also indicates that improving reliability remains important. Further testing is required to determine how the allocation behaves across additional random seeds and network configurations.

The results should therefore be interpreted within the scope of the implemented simulation and the recorded experimental settings.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This project investigated GNN-based dynamic user pairing and slot assignment for hybrid NOMA networks. A graph-based representation was used to describe users and their relationships, enabling an edge-aware GNN to generate pairing affinities and slot-preference scores. These outputs were incorporated into the allocation workflow, followed by capacity-aware scheduling and communication-performance evaluation.

The implementation included scenario generation, graph construction, GNN training, pairing, slot assignment, hybrid scheduling, and comparison with OMA, random pairing, Near-Far pairing, greedy pairing, and an optimization-based reference.

In the recorded simulation, the proposed GNN-based method achieved a mean sum rate of 17.69 Mbps, a Jain's fairness index of 0.529, and an outage percentage of 10%. The results showed that OMA achieved a higher mean sum rate, while the GNN achieved the highest fairness among the listed methods. A separate SCMA simulation demonstrated improving BER, SER, and data-rate behaviour as SNR increased.

Overall, the project demonstrates the feasibility of integrating graph-based learning into hybrid NOMA resource allocation. The results also highlight the importance of evaluating multiple performance metrics rather than relying on sum rate alone.

6.2 Future Scope

The project can be extended in several directions:

- **Larger network scenarios:** Evaluate the approach with more users, slots, and complex network topologies.
- **Realistic channel conditions:** Test the model with real wireless-channel measurements and additional fading scenarios.
- **Improved reliability:** Investigate allocation strategies that reduce outage while maintaining throughput and fairness.
- **Advanced optimization:** Explore improved scheduling and conflict-resolution techniques.

- **Generalization:** Evaluate the trained GNN on network scenarios that differ from the training distribution.
- **Computational efficiency:** Investigate inference latency and scalability for real-time resource allocation.
- **Additional baselines:** Compare with more established optimization and learning-based resource-allocation methods.

These extensions would help assess the robustness, scalability, and practical applicability of the proposed approach.

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